INCH-POUND

MIL-DTL-11713D
1 June 2018
SUPERSEDING
MIL-S-11713C
25 April 1983

DETAIL SPECIFICATION

STEEL STRIP, CARBON, SPRING: FOR MECHANICAL TIME FUZE COMPONENTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope: This specification covers carbon spring steel form (see 6.1).

1.2 Classification.

1.2.1 Composition. The steel is furnished in the following compositions, as specified (see 3.1 and 6.2).

Composition 1 - Medium carbon spring steel, low phosphorus, low sulfur.
Composition 2 - Regular medium carbon spring steel.
Composition 3 - High carbon spring steel.

1.2.2 Condition. The steel is furnished in the following conditions:

Composition 1 - Soft type annealed – dead soft.
Composition 2 - Soft type annealed – dead soft.
Composition 3 - Spheroidize annealed type.

2. APPLICABLE DOCUMENTS-

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center Philadelphia, ATTN: DSCP-NASA, 700 Robbins Avenue, Philadelphia, PA 19111-5096 or e-mail to dscpg&ispecomments@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal should apply

ASTM International Standards:

ASTM A370	- Standard Test Methods and Definitions for Mechanical Testing of Steel Products
ASTM E18	- Standard Test Methods for Rockwell Hardness of Metallic Materials.
ASTM E290	- Bend Testing of Materials

(Copies of these documents are available from http://www.astm.org or the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

SAE International

SAE AIR4127	- Steel: Chemical Composition and Hardenability
SAE AMS-STD-185	- Identification Marking of Copper and Copper Base Alloy Mill Products

(Copies of this document are available online from http://www.sae.org or the Society for Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096-001)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification should take precedence.

3. REQUIREMENTS

3.1 Pilot sample. Unless otherwise specified in the invitation for bids, contract or order see6.2), the supplier shall submit a sample, prior to the shipment of each lot, for blanking, forming, and piercing to the facility designated in the contract or order. Pilot samples from the same supplier may not be required if the accepted orders are within a three month period.

3.2 Chemical composition. The chemical composition of the steel shall meet the requirements of table I. A cast or heat analysis of each heat of steel shall be furnished by the contractor.

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TABLE I. Chemical composition, cast or heat analysis.

Composition	Carbon percent	Manganese percent	Phosphorus maximum percent	Sulfur maximum percent	Silicon percent
1	.65 - .75	.30 - .60	.025	.025	.15 - .25
2	.65 - .75	.60 - .90	.040	.050	.15 - .25
3	.90 – 1,03	.30 - ,50	.025	.025	.15 - .25

3.2.1 Product analysis tolerances. Product analysis tolerances shall be in accordance with SAE AIR4127

3.3 Mechanical properties.

3.3.1 Hardness. The steel strip shall meet the hardness requirements shown in Table II.

Table II. Hardness.

Composition	Rockwell hardness 15T scale
1	88 max.
2	88 max.
3	88 - 91

3.3.2 Bending. Steel strip of compositions 1 and 2 in the annealed condition shall withstand being bent at room tempter without cracking flat on itself longitudinally and transversely through an angle of 180 degrees around a radius equal to the thickness of the strip.

3.4 Decarburization. Depth of total and partial decarburization of the steel shall not exceed .001 inch (.025mm) or 3 percent of the thickness, whichever is greater (see 6.3).

3.5 Metallographic. Composition 3 shall show a fully spheroidized structured free from lamellar pearlite.

3.6 Dimensions and tolerances. Steel strip shall meet the dimensions specified in the invitation for bids, contract or order (see 6.2). Tolerances on specified sizes shall be as follows.

3.6.1 Thickness. Unless otherwise specified (see 6.2), the strip shall not vary in thickness more than the amounts shown in Table III.

Table III. Thickness tolerance

Ordered thickness Inch (mm)	Thickness tolerance, plus or minus Inch (mm)
Up to .012 (.30), incl.	.0003 (.0076)
Over .012 (.30) to .020 (.51) incl.	.0005 (.0127)
Over .020 (.51) to .025 (.64) incl.	.0007 (.0178)
Over .025 (.64) to .030 (.76) incl.	.0010 (.0254)

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3.6.2 Width. Unless otherwise specified (see 6.2), the strip shall not vary from the specified width by more than plus or minus .005 inch (.127 mm) for widths up to and including 9 inches (229 mm).

3.6.3 Cross-bow. Tolerances on cross-bow shall be negotiated between the contractor and the procuring activity.

3.6.4 Camber. The edgewise curvature (camber) of the strip shall not exceed .250 inch (6 mm) in any 8-foot (2.4 m) portion of total length. No reverse camber shall be permitted.

3.6.5 Coils.

3.6.5.1 Arbor diameter. The arbor diameter of coils shall be approximately 16 to 20 inches (406 to 508 mm).

3.6.5.2 Weight. Coils shall weigh approximately 200 pounds (91 kg) per 1 inch (25.4 mm) of width maximum.

3.6.5.3 Number of pieces. Coils shall contain not more than two pieces of strip, each of which shall be over 10 feet (3 m) in length.

3.7 Finish. Unless otherwise specified (see 6.2), strip shall be furnished with No. 2 (Regular Bright) finish.

3.8 Edge. Unless otherwise specified (see 6,2), strip shall be furnished with a No. 3 slit edge.

3.9 Identification marking. Each coil shall be marked in accordance with SAE AMS-STD-185.

3.10 Workmanship. The steel shall be clean and free from imperfections such as laminations, segregation, and surface defects which due to their nature, degree or extent, adversely affect the material for the intended use.

4. VERIFICATION.

4.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of not more than 1,000 pounds (450 kg) of strip of the same heat, composition, thickness and width subjected to inspection at the same time.

4.1.1 Product analysis. A lot product analysis of chemical composition shall consist of one heat.

4.2 Sampling.

4.2.1 Chemical analysis. One sample shall be taken from each lot for product analysis. Samples shall be taken in accordance with method 111 or 112 of ASTM A370.

4.2.2.1 Hardness. Four samples shall be taken from each lot. The samples shall represent each end of two different coils selected randomly.

4.2.2.2 Bending. Samples for bend testing shall be taken from two different coils in each lot. Each sample shall be of sufficient size to obtain one longitudinal and one transverse test specimen therefrom.

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4.2.2.3 Decarburization. Two samples shall be takes from each lot representing two different coils.

4.2.2.4 Metallographic. For composition 3, the samples selected for decarburization shall be used for determination of metallographic structure.

4.3 Examination.

4.3.1 Visual. A representative number of coils in each lot shall be examined for conformance with the requirements for edge (see 3.8), finish (see 3.7), identification marking (see 3.9) and workmanship (see 3.10).

4.3.2 Dimensional. A representative number of measurements shall be taken or each lot to determine conformance with the requirements for dimensions and tolerances (see 3.6) and coil (see 3.6.5)

4.3.2.1 Camber. Camber is the deviation of a side edge from a straight line. The deviation shall be obtained by placing an 8 foot (2.4 m) straight edge on the concave side and measuring the maximum distance between the steel edge and the straight edge.

4.3.3 Packaging inspection. Prior to shipment, examination shall be made to determine conformance with the requirements of section 5.

4.4 Tests.

4.4.1 Test specimens.

4.4.1.1 Chemical composition. Test samples and specimens shall be prepared in accordance with method 111 or 112 of ASTM A370, as applicable.

4.4.1.2 Hardness. Specimens shall be prepared in accordance with ASTM E18.

4.4.1.3 Bending. One longitudinal and one transverse specimen shall be prepared in accordance with ASTM E290.

4.4.1.4 Decarburization (see 6.3) and metallographic. Two specimens shall be prepared from each sample. The prepared surface of the specimen shall be not less than 1 inch (25.4 mm) in length representing the full thickness and shall be perpendicular to the direction of rolling. Specimens shall be prepared using precautions to insure freedom from disturbed metal (see 6.3).

4.4.2 Test procedures.

4.4.2.1 Chemical composition. Chemical analysis shall be conducted in accordance with method 111 or 112 of ASTM A370. Referee analysis shall be by method 111.

4.4.2.2 Hardness. Hardness tests shall be conducted in accordance with ASTM E18.

4.4.2.3 Bending. Cold bending tests shall be conducted in accordance with ASTM E290.

4.4.2.4 Decarburization (see 6.3) and metallographic. Microexamination of specimens shall be made at a magnification of 100 diameters.

4.5 Rejection.

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4.5.1 Examination. If the sampling for visual, dimensional, or packaging fails to meet the requirements of this specification when examined in accordance with 4.3, the lot shall be rejected.

4.5.2 Tests. A lot shall be rejected for failure to comply with any of the specified requirements when tested in accordance with 4.4.

4.5.3 Retests. Retests shall be permitted in accordance with ASTM A370.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended use. The steel strip covered by this specification is for a highly specialized application in timing movements and it is essential that the steel be homogeneous and uniform in thickness and quality throughout the coil.

6.1.1 Composition 1 is intended for use in mechanical time fuze escapement leavers and timing disk springs.

6.1.2 Composition 2 is for mechanical time fuze hammer springs and adjusting guides.

6.1.3 Composition 3 is for mechanical fuze disks.

6.1.4 Compositions 1 and 2 are heat treated by the purchaser. Composition 3 is used in the as received condition.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number and date of this specification.
b. Composition (see 1.2.1 and 3.2).
c. Destination of pilot sample. (see 3.1).
d. Dimensions (see 3.6).
e. Width tolerances, if other than specified (see 3.5.2).
f. Finish, if other than No.2 is required (see 3.7).
g. Edge, if other than No. 3 slit edge is required (see 3.8).
h. Lot size, if other than as specified in 4.1.
i. Levels of packaging and packing desired (see section 5).
j. When special marking for shipment requirements are necessary (see 5.2).

6.3 Decarburization. Measurement of decarburization by Microexamination is contained in Society of Automotive Engineers Recommended Practice J419, SAE Handbook 19563, Society of Automotive Engineers, 485 Lexington Avenue, New York, NY 10017

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6.4 Subject term (key word) listing.

Coil
Decarburization
Manganese
Phosphorus
Pilot sample
Spheroidized

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

MILITARY ACTIVITIES:

Custodians:
Army - MR
Navy - OS

Preparing activity:
DLA - CQ

(Project 9515-2018-001)

Reviewer:
Army - AR, MI

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